

# George Nicolas Diaz-Wahl

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## Education

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### Purdue University

*PhD candidate*

Aug 2021 – Present

*West Lafayette, IN*

### Carnegie Mellon University

*BS/MS, Thesis: The rank of modular Hecke algebras*

Aug 2017 – Mar 2021

*Pittsburgh, PA*

## Research Experience

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### IU-Purdue Joint Workshop on Prismatic F-gauges

*Purdue University in Indianapolis*

April 2026

*Indianapolis, IN*

- Presented some work in progress on understanding coherent sheaves on the prumatization and syntomification of imperfect fields

### Arizona Winter School 2024: Abelian Varieties

*University of Arizona*

Mar 2024

*Tucson, AZ*

- Rachel Pries' project group on  $p$ -rank of double covers elliptic curves
- Presented our progress on calculating the dimension of the moduli space of double covers of elliptic curves with fixed  $p$ -rank. Deduced the existence of certain supersingular singular double covers in various genera

### Honors Masters Thesis Research

*Carnegie Mellon University*

Mar 2020 - Mar 2021

*Pittsburgh, PA*

- Expository thesis on a result of my thesis advisor, Carl Wang-Erickson
- Related congruences of modular forms to rank of  $p$ -adic Hecke algebras

### Oregon State University REU

*Oregon State University*

Jun 2019 - Aug 2019

*Corvallis, OR*

- Studied (pre)periodic points of power series over Henselian fields
- Results in 2019 Oregon State REU Proceedings

## Teaching Experience

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### Graduate Teaching Assistant

*Purdue University*

Aug 2021 – Present

*West Lafayette, IN*

- MA 162 - Calculus II, Fall 2021
- Ross Fellow, Spring 2022 - Fall 2022
- MA 511 - Linear Algebra with Applications, Spring 2023
- MA 454 - Honors Abstract Algebra, Fall 2023
- MA 553 - Graduate Algebra, Fall 2023
- MA 553 - Graduate Algebra, Spring 2024
- MA 453 - Abstract Algebra, Fall 2024
- MA 108 - Mathematics as a Profession, Fall 2024, Instructor
- MA 261 - Calculus III, Fall 2025
- MA 511 - Linear Algebra with Applications, Spring 2026

### Summer Math Teacher

*Harker Upper School*

Summer 2022, 2024, 2025

*San Jose, CA*

- Jump-start to Algebra II (2025)
- Algebra I Honors (2024)

- Algebra II/Trigonometry Honors (2022)

### Summer Math Teacher

Aug 2024, Jul 2025-Aug 2025

*Art of Problem Solving Academy*

*Mountain View/San Jose, CA*

- High School Math Contests (2025)
- Middle School Math Contests (2025)
- Math Level 4 (2024)

### Mathematics Instructor

Jul 2023 – Aug 2023, Dec 2024

*Alphastar Academy*

*San Jose, CA*

- AMC 10 and 12 Courses

### Undergraduate Teaching Assistant & Grader

Aug 2018 - Mar 2021

*Carnegie Mellon University*

*Pittsburgh, PA*

- Concepts of Mathematics, 3 semesters
- Algebraic Structures, 2 semesters

### *Talks & Seminars*

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#### Seminars

Organized  $\infty$ -Categories and Higher Algebra learning seminar, Spring 2026

Organized  $p$ -adic Geometry learning seminar, Fall 2025

Organized Arithmetic student seminar, Spring 2023, Fall 2024

Coorganized Commutative Algebra student seminar, Spring 2023

Organized Moduli Spaces and Algebraic Stacks seminar, Fall 2023

#### Sample Talks

Prismatization for imperfect fields, IU-Purdue Joint Workshop on Prismatic  $F$ -gauges Prismatic

Cohomology, Learning seminar on Prismatic  $F$ -Gauges

$p$ -rank stratification in the moduli space of double covers of elliptic curves, Arizona Winter School, 2024

Étale sheaves and cohomology, Learning seminar on the Weil conjectures, 2024

The Taylor-Wiles method and patching, Learning seminar on modularity lifting, 2023

Castelnuovo-Mumford Regularity, Commutative Algebra Student Seminar, Spring 2023

Galois representations and  $(\varphi, \Gamma)$ -modules, Learning Seminar on the Emerton-Gee stack, Fall 2022

### *Honors & Awards*

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#### Excellence Award in Mathematics

*Purdue University*

*April 2026*

#### Special Award

*Purdue University*

*March 2024*

Award of \$1500 for excellence in teaching and organizer of the Moduli Spaces and Algebraic Stacks seminar.

#### Ross Fellowship

*Purdue University*

*Aug 2021 - May 2025*